

Python Functions — Practice Questions

70 questions from beginner basics to interview-level practice

Recommended focus: Start with Levels 1–3, then practice scope and interview questions.

Level 1 — Basic Functions

1. Create a function to print "Hello Python".
2. Create a function to print your name.
3. Create a function to add two numbers.
4. Create a function to subtract two numbers.
5. Create a function to multiply two numbers.
6. Create a function to divide two numbers.
7. Create a function to find the square of a number.
8. Create a function to find the cube of a number.
9. Create a function to check whether a number is even or odd.
10. Create a function to check whether a number is positive, negative, or zero.

Level 2 — Parameters & Arguments

11. Create a function that accepts two numbers and returns their sum.
12. Create a function that accepts three numbers and returns their average.
13. Create a function that accepts a number and returns its factorial.
14. Create a function that accepts a number and returns whether it is prime.
15. Create a function that accepts a number and returns whether it is a palindrome.
16. Create a function that accepts a number and returns the reverse of the number.
17. Create a function that accepts a number and returns the sum of its digits.
18. Create a function that accepts a number and returns the largest digit.
19. Create a function that accepts two numbers and returns the greater number.
20. Create a function that accepts three numbers and returns the largest number.

Level 3 — return Practice

21. Write a function that returns the factorial of a number.
22. Write a function that returns the Fibonacci series up to n terms.
23. Write a function that returns the sum of numbers from 1 to n.
24. Write a function that returns the count of digits in a number.

25. Write a function that returns the number of even digits.
26. Write a function that returns the number of odd digits.
27. Write a function that returns the reverse of a number.
28. Write a function that returns whether a number is an Armstrong number.
29. Write a function that returns whether a number is a perfect number.
30. Write a function that returns multiple results: sum, difference, and product of two numbers.

Level 4 — Functions + Loops

31. Function to print numbers from 1 to n.
32. Function to print even numbers from 1 to n.
33. Function to print odd numbers from 1 to n.
34. Function to find the sum of even numbers from 1 to n.
35. Function to find the sum of odd numbers from 1 to n.
36. Function to print the multiplication table of a number.
37. Function to count the factors of a number.
38. Function to find the sum of factors.
39. Function to check whether a number is prime.
40. Function to print all prime numbers between two numbers.

Level 5 — Scope Practice

41. Create a global variable and access it inside a function.
42. Create a local variable and try to access it outside the function.
43. Create a global variable and modify it using global.
44. Create a local variable with the same name as a global variable and observe the output.
45. Use global and compare the output with the previous question.
46. Create an outer function and inner function using nonlocal.
47. Modify an outer function's variable using nonlocal.
48. Create the same variable name in global, outer, and inner scopes and predict which value is printed.
49. Write a program demonstrating the difference between global and nonlocal.
50. Predict the output of scope-based programs without running them.

Level 6 — Interview-Level Functions

51. What happens if a function has no return statement?
52. What is the difference between print() and return?

53. What happens when return is executed inside a loop?
54. What is the difference between break and return?
55. Can a function return multiple values?
56. Why are multiple comma-separated return values stored as a tuple?
57. What is the difference between a parameter and an argument?
58. Explain positional arguments with an example.
59. Explain keyword arguments with an example.
60. Explain default arguments with an example.
61. What are *args?
62. What are **kwargs?
63. What is variable scope?
64. What is the LEGB rule?
65. What is the difference between global and nonlocal?
66. What is a nested function?
67. What is a function calling another function?
68. Can a function return another function?
69. Can a function accept another function as an argument?
70. What is recursion?

Suggested Practice Order

- 3 — Add two numbers using parameters.
- 7 — Square of a number.
- 9 — Even or odd.
- 11 — Return the sum of two numbers.
- 13 — Factorial using a function.
- 15 — Palindrome using a function.
- 17 — Sum of digits using a function.
- 21 — Factorial using return.
- 23 — Sum from 1 to n using return.
- 28 — Armstrong number using return.
- 30 — Return multiple values.
- 41 — Access a global variable inside a function.
- 43 — Modify a global variable using global.
- 46 — Use nonlocal in a nested function.
- 49 — Demonstrate global vs nonlocal.